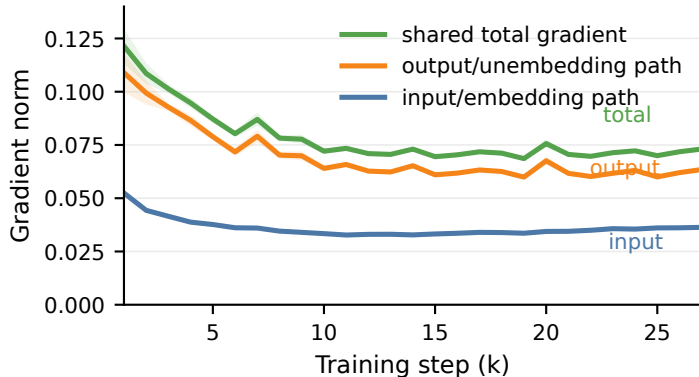
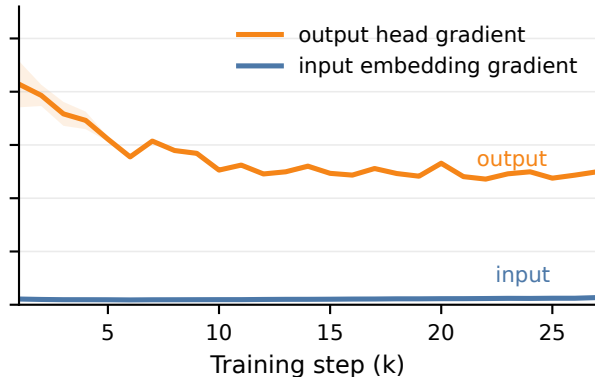


(a) S1: tied embedding/head



(b) S2: untied embedding/head



FineEdu/GPT-2 gradients averaged over seeds 42, 123, and 2026; shaded bands show ± 1 std.